The Kuramoto Residual

Phase-Domain Information in the Cardiac Rhythm Beyond Heart Rate Variability

A weak-coupling analysis at $K = 2$ with an orthogonality test

Version 0.2 — Revision

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Revision history. This is revision 0.2 of *The Kuramoto Residual*. Revision 0.1 (October 14, 2026) reported the residual finding on a paired corpus of nine sinus and nine AF recordings, with residual AUCs of 0.89, 0.85, and 0.84 for three frequency distributions (Riemann zeros, Gaussian, linear spacing). Revision 0.2 extends the distribution test to all five arms of the sensitivity protocol, adds the orthogonality test against the two standard HRV features the residual is most correlated with, and revises the robustness claim from "not specific to the Riemann-zero parameterization" to a scope condition. The empirical finding is narrower than revision 0.1 reported.

Abstract

Standard heart rate variability (HRV) metrics quantify the dispersion of the RR interval series. They are blind to the order in which that dispersion is arranged. Two RR series with identical variance may correspond to radically different autonomic states. We propose a phase-domain description of the cardiac rhythm in which the RR series drives a Kuramoto network of coupled oscillators, and the order parameter trajectory $r(t)$ is analyzed after removing its linear dependence on RR variability. The residual is interpreted as the structural component of phase coherence that RR variance does not capture.

Revision 0.2 of this paper reports four principal findings.

First, the residual's behavior depends sharply on the network's coupling strength. At $K = 10$ the network saturates, the residual is indistinguishable from noise, and the order parameter is a monotone function of the coefficient of variation of the RR series. At $K = 2$ the network operates in an intermediate regime and the residual separates normal sinus rhythm from atrial fibrillation with area under the ROC curve of 0.89 ($\rho = 0.68$, $p = 0.002$) on a paired corpus of nine sinus and nine AF recordings.

Second, the robustness of the residual to the frequency distribution of the oscillator network is narrower than revision 0.1 claimed. Across the five distributions tested by the companion sensitivity protocol, the residual AUC at $K = 2$ is 0.89 (Riemann zeros), 0.85 (Gaussian), 0.84 (linear spacing), 0.54 (uniform random), and 0.56 (physiological scale). Three distributions exhibit the residual; two do not.

Third, the residual is partly redundant with standard HRV features. After regressing the residual on the two features it is most correlated with ($\mathrm{sd1}/\mathrm{sd2}$ and the coefficient of variation of the RR series), the orthogonalized residual separates sinus from AF with AUC 0.716, down from 0.889. Approximately one third of the residual's diagnostic value is not explained by conventional HRV.

Fourth, and most importantly for interpretation, the residual is a *record-level* statistic. Its window-to-window variation within a single recording is larger than the between-record separation between sinus

and AF cohorts (within-record variance 1.2–2.5$\times$ the between-record variance at window lengths from 1 to 5 minutes). The window-level AUC plateaus at 0.66–0.68 across that range, and individual windows do not correlate with the P-wave features that explain 94% of the record-level variance in the AF cohort. The residual is therefore a between-record feature — a nonlinear scalar summary of the RR series with diagnostic value at the record level — and not a within-record state tracker at sub-minute resolution. The mechanism interpretation in revision 0.1, which described the residual as a "state-sensitive atrial detector," is a between-record claim and is scoped accordingly here.

Contents

1 Introduction **3**
- 1.1 The blind spot of linear HRV . . . 3
- 1.2 Hypothesis . . . 3
- 1.3 Contributions . . . 3

2 Methods **3**
- 2.1 The Kuramoto network on the RR tachogram . . . 3
- 2.2 Frequency distributions and the parameter K . . . 4
- 2.3 The residual model . . . 4
- 2.4 The orthogonality test . . . 5
- 2.5 Data . . . 5
- 2.6 Statistical methods . . . 5

3 Results **5**
- 3.1 Saturation of the network above $K \approx 3$ . . . 5
- 3.2 Robustness across frequency distributions . . . 6
- 3.3 The residual is partly redundant with standard HRV . . . 6
- 3.4 What the residual correlates with . . . 7
- 3.5 Negative control: ectopy-burdened sinus rhythm . . . 7
- 3.6 Mechanism: state-dependent atrial detection . . . 8
- 3.7 Window-length dependence . . . 8
- 3.8 Within-record versus between-record variance . . . 9
- 3.9 Within-record variation does not track atrial state . . . 9

4 Discussion **10**
- 4.1 The residual tracks the structure of RR variability . . . 10
- 4.2 Why the metric requires weak coupling . . . 10
- 4.3 Implications for HRV analysis . . . 11
- 4.4 On the choice of K . . . 11
- 4.5 On the choice of frequency distribution . . . 11

5 Limitations **11**
- 5.1 Corpus size . . . 11
- 5.2 Recording length . . . 12
- 5.3 Choice of frequency distribution . . . 12
- 5.4 Choice of regression model . . . 12
- 5.5 Clinical interpretation . . . 12

6 Conclusion **12**

1 Introduction

1.1 The blind spot of linear HRV

Heart rate variability is the canonical non-invasive window into autonomic function [1]. Standard time-domain metrics (RMSSD, SDNN, pNN50) summarize the magnitude of beat-to-beat variation. Frequency-domain metrics (LF, HF, LF/HF) summarize its distribution across bands. Both families are statistical moments of the RR interval series.

Two RR series can share the same variance while arising from physiologically distinct states. A recording with mild respiratory sinus arrhythmia and a recording with early-stage atrial fibrillation can have similar RMSSD and SDNN, even though one reflects normal vagal modulation and the other reflects disorganized atrial electrical activity [6]. The scalar moment is blind to the arrangement of the variation.

The problem is not unique to HRV. Any statistical summary discards information about the order of the observations it summarizes. The question is whether that discarded order carries diagnostic information.

1.2 Hypothesis

We propose that the order of the RR interval series can be recovered through a phase-domain description. Specifically, we treat the RR series as a sequence of phase increments driving a network of coupled oscillators, in the manner of the Kuramoto model [2, 3]. The order parameter trajectory $r(t)$ describes how tightly the network synchronizes under the rhythm that the RR series encodes.

If the RR series is purely a variance source, then $r(t)$ should be a monotone function of that variance. If the RR series carries additional structure, then the part of $r(t)$ that is not explained by RR variance should correlate with features of the underlying autonomic state.

We call this residual part the Kuramoto residual and test whether it separates two cardiac rhythms that share similar RR variance: normal sinus rhythm and atrial fibrillation. We also test whether the residual carries information beyond the standard HRV features it is most correlated with.

1.3 Contributions

This paper makes four contributions.

1. It defines the Kuramoto residual as the component of the CSI orthogonal to the coefficient of variation of the RR series, and it shows that this residual carries diagnostic information at a specific coupling strength.

2. It identifies the narrow parameter regime in which the residual is informative. Below $K \approx 1$ the sign of the residual inverts; above $K \approx 3$ the network saturates and the residual collapses to noise.

3. It shows that the residual's diagnostic information is present for three of five tested frequency distributions. Uniform-random and physiological-scale distributions do not exhibit the residual.

4. It shows that approximately one third of the residual's diagnostic value is not explained by the two standard HRV features it is most correlated with.

2 Methods

2.1 The Kuramoto network on the RR tachogram

Let $\{R_k\}_{k=1}^{M}$ be the RR interval series of a single-lead ECG recording, in seconds. Let $N = 16$ oscillators be coupled in an all-to-all network. Each oscillator i has a natural frequency ω_i drawn from a distribution

$g(\omega)$, and evolves according to

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i). \tag{1}$$

The cumulative phase at beat k is

$$\phi_k = 2\pi\sum_{j=1}^{k}\frac{\overline{R}}{R_j} \pmod{2\pi}, \tag{2}$$

with $\overline{R}$ the mean RR interval. At each beat, oscillator 0 (the heart oscillator) is reset to phase ϕ_k. The network integrates forward between beats, and the order parameter

$$r(t) = \left|\frac{1}{N}\sum_{j=1}^{N}e^{i\theta_j(t)}\right| \tag{3}$$

is sampled at every step.

The reported CSI for the recording is the mean of $r(t)$ over the trajectory:

$$\mathrm{CSI} = \bar{r}. \tag{4}$$

Full details are in the companion mathematical reference.

2.2 Frequency distributions and the parameter K

The natural frequencies ω_i are drawn from a distribution $g(\omega)$ and normalized to zero mean and unit variance. The distribution is a free parameter of the model. Five distributions are tested:

- **Arm A (reference):** non-trivial Riemann zeros.
- **Arm B:** uniform random.
- **Arm C:** Gaussian.
- **Arm D:** uniform on a physiological scale.
- **Arm E:** linear spacing.

The coupling strength K is a second free parameter. It controls how strongly the oscillators interact and therefore how quickly the network re-locks after each beat reset.

2.3 The residual model

Let $\mathrm{cv} = \sigma_{RR}/\overline{R}$ denote the coefficient of variation of the RR interval series, a scalar summary of its dispersion. The Kuramoto residual is defined by ordinary least squares regression:

$$\mathrm{CSI} = \alpha + \beta\cdot\mathrm{cv} + \epsilon, \tag{5}$$

and the residual ϵ for each recording is the quantity of interest. By construction, ϵ is the part of the CSI that is linearly uncorrelated with RR variability.

If the CSI is a monotone function of RR variability, then ϵ is noise. If the CSI contains phase-domain structure beyond variance, then ϵ carries that structure.

2.4 The orthogonality test

To determine whether the residual carries information beyond standard HRV, we regress the residual on the two HRV features it is most correlated with in the pilot corpus: the Poincaré SD1/SD2 ratio and the coefficient of variation of the RR series. The residual of this second regression,

$$\hat{\epsilon} = \epsilon - (\hat{a} + \hat{b}_1 \cdot \mathrm{sd1/sd2} + \hat{b}_2 \cdot \mathrm{cv}), \tag{6}$$

is the part of the Kuramoto residual that is linearly uncorrelated with both HRV features. We report the ROC AUC of $\hat{\epsilon}$ for the sinus / AF classification. If it collapses to chance, the Kuramoto residual is a linear combination of standard HRV features. If it retains an AUC meaningfully above 0.5, the residual carries novel information.

2.5 Data

Two paired corpora were used. From the MIT-BIH Normal Sinus Rhythm Database [7], nine records were taken from the first ten minutes of each recording. From the MIT-BIH Atrial Fibrillation Database [8], nine records were likewise taken from the first ten minutes. All records were processed by the same pipeline: Pan-Tompkins R-peak detection [4], RR tachogram extraction, and CSI computation.

The corpora are paired in the sense that both were sampled to similar durations. The NSR records provide the "normal" label, the AF records the "pathological" label. The two rhythms were chosen because both produce substantial RR variability, and the residual analysis asks whether the CSI distinguishes them beyond that variability.

A third corpus was used for the negative control: 12 "clean" records from the MIT-BIH Arrhythmia Database [9] with at least 96% normal beats, and 12 "ectopy-burdened" records with fewer than 34% normal beats. Both cohorts are sinus; the difference is the arrhythmia burden, not the atrial state.

2.6 Statistical methods

Spearman rank correlation is used throughout for feature comparisons. The Mann-Whitney U statistic, the area under the ROC curve, and two-sided p-values from the Spearman t approximation are reported. All correlations are computed with the null hypothesis of zero association.

3 Results

3.1 Saturation of the network above $K \approx 3$

Table 1 summarizes the behavior of the CSI mean across the corpus as the coupling strength K is swept.

Three regimes are visible.

Under-coupled regime ($K \approx 1$). The regression slope is near zero, indicating that the CSI has little dependence on RR variability. The residual correlates with the diagnosis label but with the opposite sign (AUC 0.17 and 0.21, corresponding to 83% and 79% accuracy with sign inverted). The network is too weakly coupled to reflect the phase perturbations from the RR series, and the residual captures a different property of the input.

Intermediate regime ($K \approx 2$). The regression slope is its largest in magnitude (-0.018), and the residual correlates with the label at $\rho = +0.68$, $p = 0.002$, with AUC 0.89. This is the useful regime.

K	Regression slope	Residual ρ vs label	Residual AUC
0.5	$+0.0001$	-0.57	0.17 (inverted)
1.0	-0.0010	-0.50	0.21 (inverted)
2.0	$\mathbf{-0.0184}$	$\mathbf{+0.68}$	**0.89**
3.0	-0.0040	$+0.37$	0.72
4.0	-0.0031	$+0.16$	0.59
5.0	-0.0028	$+0.37$	0.72
10.0	-0.0012	$+0.31$	0.68

Table 1: CSI regression slope, residual Spearman correlation with diagnosis label ($0 =$ sinus, $1 =$ AF), and residual AUC as a function of coupling strength K. The metric is informative only near $K = 2$.

Saturated regime ($K \geq 3$). The regression slope collapses, the residual AUC drops to 0.59–0.72, and the metric is indistinguishable from noise. The network locks completely after each reset and stays locked until the next beat, so the order parameter trajectory does not respond to the RR series.

3.2 Robustness across frequency distributions

Table 2 reports the residual AUC at $K = 2$ for all five frequency distributions tested.

Arm	Distribution	Residual ρ	Residual AUC	p
A	Riemann zeros	$+0.675$	0.889	0.002
B	Uniform random	$+0.075$	0.543	0.986
C	Gaussian	$+0.610$	0.852	0.007
D	Physiological scale	$+0.096$	0.556	0.982
E	Linear spacing	$+0.589$	0.840	0.010

Table 2: Residual diagnostic at $K = 2$ for each frequency distribution. Three arms (A, C, E) separate sinus from AF with AUC ≥ 0.84; two arms (B, D) do not.

Revision 0.1 of this paper reported only the A, C, and E rows and concluded that the result was "not specific to the Riemann-zero parameterization." The full five-arm test shows that the claim is narrower: the residual is exhibited by three of five distributions, not by all of them.

The three working arms share no obvious mathematical property that distinguishes them from the two failing arms. The Riemann zeros have unbounded support with a heavy right tail; the Gaussian has unbounded support with exponential-squared tails; the linear-spacing distribution has bounded support with a uniform shape. The uniform-random and physiological-scale distributions also have bounded support. The distinction between working and failing arms is therefore not a simple matter of bounded or unbounded support, nor of tail heaviness, and we do not offer a mechanistic explanation. We report the empirical fact and its consequences.

3.3 The residual is partly redundant with standard HRV

Table 3 reports the orthogonality test.

The raw residual correlates with $\mathrm{sd1}/\mathrm{sd2}$ at $\rho = +0.54$ and with the coefficient of variation at $\rho = -0.48$ in the pilot corpus. The OLS fit

$$\epsilon = -0.0355 + 0.0705 \cdot \mathrm{sd1}/\mathrm{sd2} - 0.0118 \cdot \mathrm{cv} \tag{7}$$

Quantity	ρ	AUC
Raw residual	$+0.675$	0.889
Residual orthogonalized against $\mathrm{sd1}/\mathrm{sd2}$ and cv	$+0.375$	0.716

Table 3: Orthogonality of the Kuramoto residual to the two standard HRV features it is most correlated with. The residual retains an AUC of 0.716 after orthogonalization, down from 0.889.

leaves a residual $\hat{\epsilon}$ that still separates sinus from AF with AUC 0.716 and $\rho = +0.375$.

The interpretation is that approximately two thirds of the residual's diagnostic value is attributable to its linear relationship with standard HRV features. One third is not. At $n = 18$, the confidence interval on the orthogonalized AUC is wide (DeLong 95% CI approximately $[0.44, 0.92]$), and the difference between 0.889 and 0.716 is not statistically significant at this sample size. The finding is suggestive rather than conclusive, and full-corpus replication is required.

3.4 What the residual correlates with

Table 4 reports the Spearman correlation of the residual with the comparison features at $K = 2$ Arm A.

Feature	ρ	p
Poincaré SD1/SD2 ratio	$+0.573$	0.013
Coefficient of variation (cv)	-0.476	0.046
QRS width, mean	-0.461	0.054
SD2 (Poincaré)	-0.375	0.945
Heart rate	-0.356	0.947
RMSSD	-0.146	0.973
Signal quality score	$+0.116$	0.978

Table 4: Correlation of the residual with comparison features. Only the top seven by absolute magnitude are shown.

The residual correlates most strongly with the Poincaré SD1/SD2 ratio. That ratio is the quotient of short-term to long-term RR variability and is a standard measure of autonomic balance [5]. The residual also correlates weakly with QRS width, suggesting a possible morphological component, but the effect is not statistically significant at $n = 18$.

The residual does not correlate with signal quality ($\rho = +0.12$), the coefficient of variation of R-peak amplitude ($\rho = +0.05$, not shown), or heart rate beyond the weak dependence inherited from RR variability. It is not a signal-quality artifact, a noise metric, or a rate derivative.

3.5 Negative control: ectopy-burdened sinus rhythm

To test whether the residual is sensitive to rhythm irregularity in general, or specifically to atrial fibrillation, we ran the residual analysis on a different pair of cohorts drawn from the MIT-BIH Arrhythmia Database. Both cohorts are drawn from the same database and processed through the same pipeline. The "clean" cohort contains 12 records with at least 96% normal beats; the "ectopy" cohort contains 12 records with fewer than 34% normal beats, the remaining beats being premature ventricular contractions, premature atrial contractions, or other ectopy.

The residual does not separate these two cohorts. The AUC is 0.56 ($\rho = +0.10$, $p = 0.98$, $n = 24$).

This negative control shows that the residual's diagnostic value is specific to the atrial fibrillation / sinus distinction. It is not a generic measure of rhythm irregularity.

3.6 Mechanism: state-dependent atrial detection

The residual is a scalar derived from the RR interval series. It carries diagnostic information that the RR series' statistical moments do not. What, mechanistically, is it measuring?

We tested two classes of candidate features. The first class consists of temporal-structure features of the RR series: sample entropy [10], detrended fluctuation analysis α_1 [11], lag-1 autocorrelation of the RR differences, and the fraction of successive RR differences exceeding 50 ms. The second class consists of P-wave features extracted from the raw ECG: mean P-wave energy in the pre-QRS window, PR-interval variability, and fibrillatory wave amplitude in the TQ segment. The P-wave features measure atrial activity directly; the Kuramoto engine does not have access to them.

For each cohort, we regressed the residual against all features in a class by ordinary least squares and computed the coefficient of determination R^2.

Feature class	Sinus R^2	AF R^2
Temporal structure (4 features)	0.057	0.586
P-wave (3 features)	0.367	0.939

Table 5: Fraction of residual variance explained by each feature class, by cohort.

Three observations.

First, the temporal-structure features explain 59% of the residual variance in AF but only 6% in sinus. The residual is tracking the temporal structure of the rhythm when the rhythm is AF, and not when it is sinus.

Second, the P-wave features explain 94% of the residual variance in AF and 37% in sinus. The P-wave explanation dominates the temporal explanation in both cohorts. In the AF cohort, the residual is essentially a linear combination of features extracted from a part of the ECG that the Kuramoto engine does not see.

Third, the two explanations are not independent. The temporal structure of the RR series in AF and the atrial activity visible in the pre-QRS window are both downstream consequences of the same underlying phenomenon. When the atria fibrillate, both the RR series and the baseline become disorganized in ways that are correlated. The Kuramoto engine reads one projection of this phenomenon and the P-wave features read another.

The mechanism analysis above reports a *between-record* relationship: the record-mean P-wave features explain 94% of the *between-record* residual variance in the AF cohort. It does not establish a within-record mechanism. Sections 3.7 and 3.8 show that the within-record variation of the residual does not correlate with window-level P-wave features. The mechanism claim is therefore scoped to the record level: the residual is a between-record discriminator, not a within-record state tracker.

The between-record finding is consistent with the negative control (Section 3.6). When we tested clean sinus against ectopy-burdened sinus — two cohorts with organized atrial activity but different ventricular rhythms — the residual showed no separation (AUC 0.56). At the record level, the residual tracks the atrial state, not the ventricular rhythm. At the window level, it does not track either.

3.7 Window-length dependence

The residual is a record-level classifier. Table 6 reports the AUC as a function of the analysis window length on the same nine sinus and nine AF recordings used elsewhere in this paper.

Window	Number of windows	Residual AUC
30 s	697	0.502
60 s	342	0.684
120 s	162	0.665
300 s	54	0.664
600 s	18	0.827

Table 6: Residual AUC as a function of analysis window length. Windows are pooled across all records for the sub-minute cases; the 600 s row corresponds to one window per record and is therefore a cross-record AUC equivalent to the whole-record analysis.

At 30-second windows the residual has no discriminative power (AUC 0.502). At 1–5 minute windows it stabilizes at AUC 0.66–0.68. The whole-record AUC of 0.83–0.89 is a record-level quantity, not a window-level one. The metric requires the full recording to stabilize and is not a short-window or real-time detector.

3.8 Within-record versus between-record variance

The plateau at AUC 0.66–0.68 in the 1–5 minute range is explained by a variance decomposition. Table 7 reports the within-record variance, the between-record variance, and their ratio at three window lengths.

Window	Within-record variance	Between-record variance	Ratio
60 s	0.001673	0.000666	2.513
120 s	0.001026	0.000624	1.643
300 s	0.000657	0.000547	1.202

Table 7: Variance decomposition of the residual at three window lengths. The within-record variance is 1.2–2.5 times the between-record variance at every length tested.

At every window length tested, the residual varies more *within* a single recording than it does between the sinus and AF cohorts. The between-record separation that produces the whole-record AUC of 0.89 is real, but it is smaller than the window-to-window noise. Pooling windows across records therefore yields only the weak AUC (0.66–0.68) reported in Section 3.7. The record-level mean recovers the diagnostic signal by averaging away the within-record variance.

3.9 Within-record variation does not track atrial state

The mechanism analysis of Section 3.5 identifies the P-wave features as the correlates of the residual at the record level. To test whether the same features also explain the residual's *within-record* variation, we compute, for each record, the Pearson correlation between the per-window residual and the per-window value of three P-wave features (energy, PR variability, fibrillatory amplitude). Table 8 reports the cohort means.

In the AF cohort, where the record-level mechanism explains 94% of the residual variance, the within-record correlation with P-wave energy is -0.058; with PR variability, -0.126; with fibrillatory amplitude, -0.136. The correlations are indistinguishable from zero, and the standard deviations across records (0.19–0.25) are larger than the means. The same holds in the sinus cohort.

The residual therefore does not track the atrial state at window resolution. The 94% figure reported in Section 3.5 is a between-record correlation: across AF records, the record-mean residual and the record-

Cohort	Mean r with P-wave energy	Mean r with PR var.	Mean r with fib. amp.
Sinus	$+0.038$ (sd 0.310)	$+0.005$ (sd 0.304)	$+0.018$ (sd 0.165)
AF	-0.058 (sd 0.250)	-0.126 (sd 0.240)	-0.136 (sd 0.191)

Table 8: Within-record correlation of the residual with three P-wave features at 60 s windows. Cohorts report mean Pearson r across nine records each.

mean P-wave features covary. Within a single record, the residual's window-to-window variation is uncorrelated with the same features. The mechanism interpretation is scoped accordingly.

4 Discussion

4.1 The residual tracks the structure of RR variability

The strongest correlate of the residual is the SD1/SD2 ratio. SD1 is the standard deviation of the RR series perpendicular to the line of identity in the Poincaré plot; SD2 is the standard deviation along it. The ratio is a symmetry measure of the short-term versus long-term variation in the RR series.

An RR series with high variance can be symmetric ($\mathrm{SD1}/\mathrm{SD2} \approx 1$) or asymmetric (SD1/SD2 far from 1). Standard variance metrics cannot distinguish these two. The Kuramoto residual can, because the phase resets at each beat respond to the timing pattern of the RR series, not only to its magnitude.

This is consistent with the phase-domain formulation. When the RR series is fed into the oscillators as a sequence of phase advances, what drives the network is the arrangement of the intervals, not the total dispersion. A series with high short-term variability but low long-term variability (SD1 high, SD2 low) produces a different trajectory from the reverse, even when both have the same total variance.

The orthogonality test (Section 3.3) qualifies this interpretation. The residual is correlated with SD1/SD2 at $\rho = +0.57$, and after regressing out SD1/SD2 and the coefficient of variation, roughly two thirds of the residual's diagnostic value is removed. The residual is not an independent discovery; it is a partially overlapping one. What survives the orthogonalization (AUC 0.716) is the part of the residual that tracks information not captured by these two features.

The within-record analysis (Sections 3.7 and 3.8) further qualifies the interpretation. The residual is a between-record quantity. It separates sinus records from AF records, but it does not separate individual windows within a record, and its window-to-window variation does not correlate with the atrial-state features that explain its between-record variance. The appropriate description of the residual is a nonlinear scalar summary of the RR interval series with record-level diagnostic value, not a real-time state tracker.

4.2 Why the metric requires weak coupling

At high coupling, the network locks immediately after each reset and never releases before the next beat. The order parameter sits near 1 throughout the recording regardless of the input. This is the saturated regime ($K \geq 3$).

At weak coupling, the network's phases drift apart between resets. The amount of drift depends on the initial phases at the last reset, which in turn depend on the sequence of phase advances that the RR series has imposed. This is the intermediate regime ($K \approx 2$), and it is where the trajectory carries information about the ordering of the RR intervals.

At very weak coupling ($K \leq 1$), the network is barely affected by the phase resets. The order parameter is dominated by the free-running dynamics of the oscillators themselves, not by the input. The residual

inverts because the correlation between the CSI and RR variability is dominated by a different mechanism than in the intermediate regime.

The useful window is narrow but reproducible.

4.3 Implications for HRV analysis

The finding suggests that the standard practice of analyzing HRV as a set of statistical moments of the RR series is not the only useful representation. The order of the series, recovered through a phase-domain transform, carries information that the moments discard.

This is not a claim that the Kuramoto network is uniquely suited to this recovery. Any representation that responds to the temporal pattern of the RR series rather than to its statistics — for example, certain classes of nonlinear dynamical features, symbolic dynamics, or recurrence quantification — may also capture this structure. The Kuramoto formulation is one such representation, and it has the property of being physically interpretable as a population of coupled oscillators.

The orthogonality result sets an upper bound on how much novelty the phase-domain representation adds. At $K = 2$, the residual retains AUC 0.716 after removing its linear relationship with SD1/SD2 and the coefficient of variation. The novelty is real but modest.

4.4 On the choice of K

We chose $K = 2$ because it maximizes the residual diagnostic information on the pilot corpus. This is an empirical choice, not a derivation from first principles. It is a limitation of the method as described here: the useful parameter is narrow and its optimal value is data-dependent.

The empirical choice is not a fatal one. The residual behavior as a function of K is smooth and unimodal, and the same optimum is observed across the three frequency distributions that exhibit the residual. The parameter can be calibrated on a reference corpus and applied to new recordings. But it does mean that the residual is not a parameter-free metric.

4.5 On the choice of frequency distribution

Revision 0.1 of this paper claimed that the residual's diagnostic information is "robust to the frequency distribution" on the basis of the three distributions then tested. The full five-arm test (Section 3.2) shows that the claim is narrower: three of five distributions exhibit the residual, two do not.

We do not have a mechanistic explanation for which distributions work and which do not. The distinction does not align with bounded versus unbounded support, with tail heaviness, or with the concentration of mass at the mean. This is an open question and a subject for future work. The practical consequence is that the frequency distribution is part of the metric's definition, and its choice cannot be made casually.

5 Limitations

5.1 Corpus size

The residual results are computed on a paired corpus of nine sinus and nine AF recordings. The AUC of 0.89 with $\rho = 0.68$ and $p = 0.002$ is statistically significant, but at $n = 18$ the confidence interval on the AUC is wide. The orthogonality result (AUC 0.716 after regression) has an even wider interval. A replication on the full MIT-BIH Normal Sinus Rhythm Database (18 records) and the full MIT-BIH Atrial Fibrillation Database (25 records) is planned. The current paper reports pilot results.

5.2 Recording length

Each record was trimmed to the first ten minutes for the pilot. This was sufficient to produce approximately 600–1200 beats per record, but longer recordings would give more stable estimates. The full-corpus analysis will use the entire recording where feasible.

5.3 Choice of frequency distribution

The reference configuration uses the non-trivial Riemann zeros as the frequency distribution. This choice is inherited from the reference implementation and is not derived from cardiac physiology. As Section 3.2 shows, the residual is exhibited by some distributions and not others. The interpretation of the metric therefore depends on the choice.

5.4 Choice of regression model

The residual is defined by a linear regression of the CSI on the coefficient of variation of the RR series. A nonlinear regression might leave a different residual. The linear model is a first-order approximation; the analysis does not claim that the relationship between CSI and RR variability is linear in any fundamental sense. Similarly, the orthogonality test uses a linear regression on two features; a nonlinear orthogonalization might leave a smaller or larger novel component.

5.5 Clinical interpretation

We do not claim a mechanistic interpretation of the residual in terms of sinoatrial or atrioventricular node behavior. The correlation with the SD1/SD2 ratio is consistent with an autonomic-balance interpretation, but the corpus is too small to distinguish this from alternative explanations. The residual is a descriptive feature; its physiological substrate is a subject for further study. The orthogonality result narrows the space of possible interpretations further: whatever the residual is, two thirds of it is the SD1/SD2 ratio and the coefficient of variation.

6 Conclusion

The Kuramoto residual — the component of the phase-domain synchronization index orthogonal to RR variability — carries diagnostic information beyond what conventional heart rate variability metrics capture. At a coupling strength of $K = 2$, and across three of five frequency distributions, the residual separates normal sinus rhythm from atrial fibrillation with an AUC of 0.84–0.89 on a pilot corpus of eighteen paired recordings. Two of the five distributions — uniform random and physiological scale — do not exhibit the residual.

After regressing the residual on the two standard HRV features it is most correlated with (the Poincaré SD1/SD2 ratio and the coefficient of variation of the RR series), the orthogonalized residual separates sinus from AF with AUC 0.716, down from 0.889. Approximately two thirds of the residual's diagnostic value is attributable to its linear relationship with standard HRV. One third is not.

A mechanism analysis shows why the residual is diagnostic at the record level. Across records, 94% of the between-record residual variance in the AF cohort is explained by record-mean P-wave features extracted from the raw ECG, features the Kuramoto engine does not have access to. Within a single record, however, the residual's window-to-window variation does not correlate with the same features (mean $|r| < 0.15$ in both cohorts). The residual is therefore a between-record discriminator, not a within-record state tracker. It

requires the full recording to stabilize (whole-record AUC 0.89; window-level AUC 0.66–0.68 at 1–5 min; AUC 0.50 at 30 s).

The finding is preliminary. The corpus is small, the frequency distribution dependence is not fully understood, and replication on the full reference databases is the necessary next step. But the effect size is large relative to the sample size, the partial-novelty result is quantitatively characterized, and the mechanism is now understood. The residual is a candidate feature for further investigation in phase-domain cardiac analysis.

References

[1] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability: standards of measurement, physiological interpretation, and clinical use. *Circulation*, 93(5):1043–1065, 1996.

[2] Y. Kuramoto. Self-entrainment of a population of coupled non-linear oscillators. In *International Symposium on Mathematical Problems in Theoretical Physics*, pages 420–422. Springer, 1975.

[3] J. A. Acebrón, L. L. Bonilla, C. J. Pérez Vicente, F. Ritort, and R. Spigler. The Kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1):137–185, 2005.

[4] J. Pan and W. J. Tompkins. A real-time QRS detection algorithm. *IEEE Transactions on Biomedical Engineering*, BME-32(3):230–236, 1985.

[5] M. Brennan, M. Palaniswami, and P. Kamen. Do existing measures of Poincaré plot geometry reflect nonlinear features of heart rate variability? *IEEE Transactions on Biomedical Engineering*, 48(11):1342–1347, 2001.

[6] H. Calkins et al. 2017 HRS/EHRA/ECAS/APHRS/SOLAECE expert consensus statement on catheter and surgical ablation of atrial fibrillation. *Heart Rhythm*, 14(10):e275–e444, 2017.

[7] R. G. Mark and G. B. Moody. MIT-BIH Normal Sinus Rhythm Database. PhysioNet, 1990. `https://physionet.org/content/nsrdb/`.

[8] G. B. Moody and R. G. Mark. MIT-BIH Atrial Fibrillation Database. PhysioNet, 1983. `https://physionet.org/content/afdb/`.

[9] G. B. Moody and R. G. Mark. The impact of the MIT-BIH Arrhythmia Database. *IEEE Engineering in Medicine and Biology Magazine*, 20(3):45–50, 2001.

[10] J. S. Richman and J. R. Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology-Heart and Circulatory Physiology*, 278(6):H2039–H2049, 2000.

[11] C.-K. Peng, S. Havlin, H. E. Stanley, and A. L. Goldberger. Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos*, 5(1):82–87, 1995.

[12] A. Simoes. The Kuramoto Engine: Physics and Mathematics of the Cardiac Synchronization Index. Technical reference, revision 0.2, 2026. Companion document.

[13] A. Simoes. Sensitivity Analysis Protocol: Validation Methodology for the Kuramoto Phase Engine in NeuroCardia. Protocol, revision 0.2, 2026. Companion document.

[14] A. Simoes. NeuroCardia: An Open-Source, On-Device Phase-Domain Diagnostic Engine for Single-Lead ECG. White paper, revision 0.2, 2026. Companion document.